2 Real Analysis 2023
Tutorial 10

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Compact sets in $\mathbb{R}$

Problem 39 Decide whether the following sets are compact. Give a justification.

(a) $\mathbb{N}$,

(b) $\mathbb{R}$,

(c) $A = \{x \in [0, 10] : x \neq 5\}$,

(d) $B = \{x \in \mathbb{R} : 1 \leq |x| \leq 2\}$,

(e) $C = \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$,

(f) $D = \left\{\frac{1}{n} : n \in \mathbb{N}\right\} \cup \{0\}$.

Solution.

Recall that a subset of $\mathbb{R}$ is compact if and only if it is closed and bounded.

(a) $\mathbb{N}$ is not compact, because $\mathbb{N}$ is unbounded.

(b) $\mathbb{R}$ is not compact, because $\mathbb{R}$ is unbounded.

(c) Notice that $A = [0, 5) \cup (5, 10]$. A is not compact because A is not closed: it does not contain its limit point 5.

(d) $B = [-2, -1] \cup [1, 2]$ is compact. This follows from the fact that B is bounded (for example, -2 is a lower bound and 2 is an upper bound for B) and closed (it is a union of two closed intervals).

(e) C is not compact because it is not closed: it does not contain its limit point 0.

(f) D is compact. Indeed, it is bounded (for example, 0 is a lower bound and 1 is an upper bound for D) and closed — this follows from the fact that the only limit point of D is 0 (see Problem 35).

Limits of functions

Problem 40 Prove by definition of the limit:

(a) If $f(x) = c$ with some $c \in \mathbb{R}$ for all $x \in A$, and if a is a limit point of A, then $\lim_{x\to a} f(x) = c$.

(b) If $f(x) = x$ for all $x \in A$, and if a is a limit point of A, then $\lim_{x\to a} f(x) = a$.

Solution.

(a) Notice that $|f(x) - c| = 0$ for any $x \in A$. Given $\varepsilon > 0$, we can take any $\delta > 0$ to obtain
$$0 < |x - a| < \delta \implies |f(x) - c| < \varepsilon.$$
Thus, $\lim_{x\to a} f(x) = c$ by definition.

(b) Notice that $|f(x) - a| = |x - a|$ for any $x \in A$. Given $\varepsilon > 0$, we can take any $\delta = \varepsilon$ to obtain
$$0 < |x - a| < \delta \implies |f(x) - a| < \varepsilon.$$
Thus, $\lim_{x\to a} f(x) = a$ by definition.

Problem 41 Let $f : \mathbb{R} \to \mathbb{R}$ and $a, L \in \mathbb{R}$. Show that $\lim_{x\to a} f(x) = L$ if and only if $\lim_{h\to 0} f(a + h) = L$.

Solution.

Introduce the function $F(h) = f(a + h)$, $h \in \mathbb{R}$. We have to prove that $\lim_{x\to a} f(x) = L$ if and only if $\lim_{h\to 0} F(h) = L$.

Let $\lim_{x\to a} f(x) = L$. This is equivalent to
$$\forall\, \varepsilon > 0\ \exists\, \delta > 0 : 0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon.$$
Take $h = x - a$, then $x = a + h$ and $f(x) = F(h)$. Substituting these equations in the above formula, we obtain the equivalent formula
$$\forall\, \varepsilon > 0\ \exists\, \delta > 0 : 0 < |h| < \delta \implies |F(h) - L| < \varepsilon.$$
This is equivalent to $\lim_{h\to 0} F(h) = L$.

Problem 42 Show that $\lim_{x\to 0} \cos\left(\frac{1}{x}\right)$ does not exist, but $\lim_{x\to 0} x\cos\left(\frac{1}{x}\right) = 0$.

Solution.

We first show that $\lim_{x\to 0} \cos\left(\frac{1}{x}\right)$ does not exist. Notice that $\cos\left(\frac{1}{x}\right) = 1$ if and only if $\frac{1}{x} = 2\pi k$, $k \in \mathbb{Z}$. Take $x_n = \frac{1}{2\pi n}$, $n \in \mathbb{N}$. We have $\lim_{n\to\infty} x_n = 0$ and $\lim_{n\to\infty} \cos(x_n) = \lim_{n\to\infty} 1 = 1$.

On the other hand, $\cos\left(\frac{1}{x}\right) = 0$ if and only if $\frac{1}{x} = \frac{\pi}{2} + \pi k$, $k \in \mathbb{Z}$. Take $y_n = \frac{1}{\frac{\pi}{2}+\pi n}$, $n \in \mathbb{N}$. Then $\lim_{n\to\infty} y_n = 0$ and $\lim_{n\to\infty} \cos(y_n) = \lim_{n\to\infty} 0 = 0$. Thus, we found two sequences $(x_n)_{n=1}^{\infty}$, $(y_n)_{n=1}^{\infty}$ such that $\lim_{n\to\infty} x_n = \lim_{n\to\infty} y_n = 0$, but $\lim_{n\to\infty} \cos(x_n) \neq \lim_{n\to\infty} \cos(y_n)$. This shows that $\lim_{x\to 0} \cos\left(\frac{1}{x}\right)$ does not exist.

Now we show that $\lim_{x\to 0} x\cos\left(\frac{1}{x}\right) = 0$. Given $\varepsilon > 0$, take $\delta = \varepsilon > 0$. Then

$$0 < |x| < \delta = \varepsilon \implies \left|x\cos\left(\frac{1}{x}\right) - 0\right| = |x|\left|\cos\left(\frac{1}{x}\right)\right| \leq |x| < \varepsilon.$$

Thus, by definition, $\lim_{x\to 0} x\cos\left(\frac{1}{x}\right) = 0$.

Problem 43 Let $f : \mathbb{R} \to \mathbb{R}$ be a function with $\lim_{x\to 3} f(x) = 2$. Prove by the definition of the limit that

$$\text{(a)}\ \lim_{x\to 3}(f(x)+1)^2 = 9, \qquad \text{(b)}\ \lim_{x\to 3} \frac{2f(x)-3}{4f(x)-9} = -1.$$

Solution.

Notice that $\lim_{x\to 3} f(x) = 2$ means, by definition, that

$$\forall\, \varepsilon > 0\ \exists\ \delta_\varepsilon > 0 : 0 < |x-3| < \delta_\varepsilon \implies |f(x) - 2| < \varepsilon.$$

We will use this notation for δ_ε below.

(a) We have

$$|(f(x)+1)^2 - 9| = |f(x)^2 + 2f(x) - 8| = |f(x) - 2| \cdot |f(x) + 4|.$$

With $\delta_1 > 0$ as above we have

$$0 < |x-3| < \delta_1 \implies |f(x) - 2| < 1.$$

For such x we have $1 < f(x) < 3$, and hence $|f(x) + 4| < 3 + 4 = 7$. Therefore,

$$0 < |x-3| < \delta_1 \implies |(f(x)+1)^2 - 9| = |f(x)-2| \cdot |f(x)+4| < 7|f(x) - 2|.$$

Given $\varepsilon > 0$, there exists $\delta_{\varepsilon/7} > 0$ with

$$0 < |x-3| < \delta_{\varepsilon/7} \implies |f(x) - 2| < \frac{\varepsilon}{7}.$$

Put $\delta = \min\{\delta_1, \delta_{\varepsilon/7}\} > 0$. Then

$$0 < |x-3| < \delta \implies |(f(x)+1)^2 - 9| < 7|f(x)-2| < 7 \cdot \frac{\varepsilon}{7} = \varepsilon$$

as desired. This proves that $\lim_{x\to 3}(f(x)+1)^2 = 9$.

(b) We have

$$\left|\frac{2f(x)-3}{4f(x)-9}+1\right| = \left|\frac{2f(x)-3+4f(x)-9}{4f(x)-9}\right| = \left|\frac{6f(x)-12}{4f(x)-9}\right| = \frac{6|f(x)-2|}{|4f(x)-9|}.$$

Now we have to bound $|4f(x)-9|$ from below. If x is close to 3, then $f(x)$ is close to 2 and the term $4f(x)-9$ is negative. Let us see how we can get the estimate $-\frac{3}{2} < 4f(x)-9 < -\frac{1}{2}$. This is equivalent to $\frac{15}{2} < 4f(x) < \frac{17}{2}$, then to $\frac{15}{8} < f(x) < \frac{17}{8}$ and finally to $-\frac{1}{8} < f(x)-2 < \frac{1}{8}$. Thus, we can take $\varepsilon = \frac{1}{8}$. We know that there exists $\delta_{1/8} > 0$ such that

$$0 < |x-3| < \delta_{1/8} \implies |f(x)-2| < \frac{1}{8}.$$

For $0 < |x-3| < \delta_{1/8}$ we have $|4f(x)-9| > \frac{1}{2}$, hence

$$\left|\frac{2f(x)-3}{4f(x)-9}+1\right| = \frac{6|f(x)-2|}{|4f(x)-9|} < \frac{6|f(x)-2|}{1/2} = 12|f(x)-2|.$$

Given $\varepsilon > 0$, there exists $\delta_{\varepsilon/12} > 0$ such that

$$0 < |x-3| < \delta_{\varepsilon/12} \implies |f(x)-2| < \frac{\varepsilon}{12}.$$

Put $\delta = \min\{\delta_{1/8}, \delta_{\varepsilon/12}\} > 0$. Then

$$0 < |x-3| < \delta \implies \left|\frac{2f(x)-3}{4f(x)-9}+1\right| < 12|f(x)-2| < 12\cdot\frac{\varepsilon}{12} = \varepsilon$$

as desired. This proves that $\lim_{x\to 3}\frac{2f(x)-3}{4f(x)-9} = -1$.